

Solution Requirements

Date	29 June 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviours, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviours (How the system should *be*).

Step 1: Functional Requirements (FR)

S. No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	No user authentication is implemented. The application is publicly accessible through the home page.	Low
2	Authorization levels	No user roles or permission levels are implemented. All users have equal access to prediction functionality.	Low
3	External interfaces	The system loads a trained Machine Learning model (model.pkl) using the Pickle library. It uses Flask for the web interface and NumPy for data processing.	High

4	Transactions processing	Users enter Life Expectancy, Mean Schooling, Expected Schooling, and GNI. The application validates inputs, sends them to the trained model, predicts the HDI score, classifies the result, and displays the output.	High
5	Reporting	The system displays the predicted HDI value and its Human Development category on the result page.	High
6	Business rules, etc.	HDI is classified based on prediction value: $\geq 0.80 \rightarrow$ Very High, $\geq 0.70 \rightarrow$ High, $\geq 0.55 \rightarrow$ Medium, otherwise Low Human Development.	High
7	Compliance to laws or regulations	The system does not store personal user information. Basic data privacy is maintained as no user data is permanently saved.	Medium
8	<i>Other</i>	The application handles invalid inputs using exception handling and displays an error message without crashing the application.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should generate predictions quickly after user input.	Prediction completed in less than 2 seconds .
2	Scalability	The system should support multiple prediction requests without affecting performance.	Supports multiple concurrent users on a Flask server.
3	Security & Data Privacy	No sensitive information is stored. User input is processed only during prediction and exception handling prevents application crashes.	No permanent storage of user data; secure server-side processing
4	Reliability & Availability	The application should remain operational whenever the Flask server is running.	99% availability during server runtime.
5	Usability & Accessibility	The interface should be simple, user-friendly, and easy to navigate with clearly labelled input fields and prediction results.	Users can complete a prediction in less than 1 minute .
6	Other	The application is modular, with separate files for templates, static resources, the trained model, and Flask logic, making future updates easier.	Easy maintenance and code updates without affecting core functionality.